Radboud University Nijmegen

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The Prosciutto Question

Bachelor Thesis in Mathematics

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Chapter 1

Preliminaries

1.1 General knowledge and notation

We begin by recalling and defining some general concepts that play an important role throughout this thesis.

Let X be a Banach space and $T > 0$. A function $f : [0,T] \to X$ is called *strongly measurable* if there is a sequence of simple functions $(f_n)_{n\in\mathbb{N}}$ that converges pointwise to f at almost every $t \in [0,T]$. Now for $p \in [1,\infty]$, the space $L^p(0,T;X)$ consists of all strongly measurable functions u such that

$$\|u\|_{L^p(0,T;X)} := \left(\int_0^T \|u(t)\|_X \, dt \right)^{1/p} < \infty$$

for $p \in [1,\infty)$, and

$$\|u\|_{L^\infty(0,T;X)} := \operatorname*{ess\,sup}_{t\,\in\,[0,T]} \|u(t)\|_X < \infty.$$

Lastly, $C([0,T];X)$ denotes the space of all continuous functions $u : [0,T] \to X$ with

$$\|u\|_{C([0,T];X)} := \max_{t\in[0,T]} \|u(t)\|_X < \infty.$$

We can take X to be, for instance, $L^2(\Omega)$. Then formally, $C([0,T];L^2(\Omega))$ consists of continuous functions u from the interval time [0, T] to $L^2(\Omega)$. For a more intuitive interpretation, you might feel inclined to interpret them as the corresponding functions $\widetilde{u}$ from $[0,T] \times \Omega$ to $\mathbb{R}$, given by $\widetilde{u}(t,x) = u(t)(x)$. We do need to be careful when we use this interpretation, though, as the notation might make it seem like these functions $\widetilde{u}$ are continuous in time, while this is generally not the case. A more accurate way to think about this, is that at every time t, the set of points $x \in \Omega$ where $\widetilde{u}$ has a discontinuity in time has measure zero.

The next ideas are often used to translate local properties into global results. This method plays a fundamental part in our later discussions about the formulation and analysis of weak forms and solutions to PDEs.

Let X now be a topological space. A function $\phi : X \to \mathbb{R}$ is said to have *compact support* if the set

$$\operatorname{supp}(\phi) := \overline{\{x \in X \mid \phi(x) \neq 0\}}$$

is a compact subset of X. For any open $\Omega \subseteq \mathbb{R}^m$, the space of all smooth functions with compact support in Ω is denoted by $C_c^\infty(\Omega)$.

An important property of these functions is that they must be 0 on the boundary of Ω. In the particular context of PDE analysis, they are often referred to as *test functions*, but a more general term is *variations*. They have a wide range of applications, discussed in the field of the Calculus of Variations. A large portion of its theory is built around the following result, hence its name.

Lemma 1.1 (Fundamental Lemma of the Calculus of Variations)
Let $\Omega \subseteq \mathbb{R}^m$ be open and $f \in L^2(\Omega)$. If

$$\int_\Omega f\phi = 0$$

holds for all $\phi \in C_c^\infty(\Omega)$, then $f = 0$ almost everywhere in Ω.

Multiplying with a test function and integrating is sometimes referred to as *testing against* a test function $\phi \in C_c^\infty(\Omega)$.

Definition 1.2
When we are differentiating functions in C^k defined in a subset of $\mathbb{R}^m$, we sometimes use the abbreviated notation

$$D^\alpha := \frac{\partial^{|\alpha|}}{\partial x_1^{\alpha_1} \dots \partial x_m^{\alpha_m}},$$

where $\alpha = (\alpha_1, \dots, \alpha_m) \in \mathbb{N}^m$ is called a multi-index and $|\alpha| := \alpha_1 + \cdots + \alpha_m \leq k$.

Definition 1.3
Let $\Omega \subseteq \mathbb{R}^m$ be open. A function $f : \Omega \to \mathbb{R}$ is said to be *locally integrable* if

$$\int_K |f| < \infty$$

for all compact subsets K of Ω. The space of all locally integrable functions on Ω is denoted by $L^1_{\text{loc}}(\Omega)$.

An important fact that should be noted is that every funcion that is square-integrable is also locally integrable. That is, $L^2(\Omega) \subseteq L^1_{\text{loc}}(\Omega)$. For the sake of completeness, we will sometimes state definitions or results with the hypothesis that a function is locally integrable, only to use them on functions that are square-integrable.

1.2 Weak derivatives and Sobolev spaces

Before we talk about weak solutions to PDEs, we need to define spaces of functions that are potential candidates for solutions. We derive this space by weakening the conditions of differentiable functions.

Consider an open and bounded domain $\Omega \subseteq \mathbb{R}^m$. Then for any continuously differentiable function $u \in C^1(\Omega)$ and test function $\phi \in C_c^\infty(\Omega)$, we have

$$\int_\Omega u_{x_i}\phi = \int_{\partial\Omega} \frac{\partial u}{\partial \nu}\phi \, dS - \int_\Omega u\phi_{x_i} = -\int_\Omega u\phi_{x_i}, \tag{1.1}$$

since ϕ is 0 on the boundary. Consequentially, for any function $u \in C^k(\Omega)$ and multi-index $\alpha \in \mathbb{N}^m$ with $|\alpha| \leq k$, we have

$$\int_\Omega D^\alpha u\phi = (-1)^{|\alpha|}\int_\Omega uD^\alpha\phi. \tag{1.2}$$

Note that for this last expression to make sense, the function $D^\alpha u$ does not necessarily have to be continuous. In fact, we can completely remove the hypothesis that u is differentiable, and define the *weak derivative* of u as a function that behaves in a similar way as eq. (1.2) when tested against a function $\phi \in C_c^\infty(\Omega)$.

Definition 1.4
Let $u, v \in L^1_{\text{loc}}(\Omega)$ and $\alpha \in \mathbb{N}^m$. We say that v is the *α-th weak derivative* of u if

$$\int_\Omega v\phi = (-1)^{|\alpha|}\int_\Omega uD^\alpha\phi \tag{1.3}$$

for every test function $\phi \in C_c^\infty(\Omega)$.

We often use the same notation for weak derivatives as we do for regular derivatives. When $D^\alpha u$ is used, we should carefully think about which derivative is meant here.

This notion essentially has the same idea of regular derivatives, but the requirements on u are much weaker. To be able to talk about *the* α-th weak derivative, we show that weak derivatives are unique in L^1, or unique in measure.

Proposition 1.5
The α-th weak derivative of u, if it exists, is uniquely defined up to a set of measure zero.

Proof. Suppose that v and w are two α-th weak derivatives of u. Then for every $\phi \in C_c^\infty(\Omega)$,

$$\int_\Omega v\phi = (-1)^{|\alpha|}\int_\Omega uD^\alpha\phi = \int_\Omega w\phi,$$

which gives

$$\int_\Omega (v-w)\phi = 0$$

for every $\phi \in C_c^\infty(\Omega)$. We can now use the Fundamental Lemma of the Calculus of Variations (FLCV) to see that $v-w$ is 0 almost everywhere, which means that $u = v$ almost everywhere, completing the proof. □

We now define the *Sobolev space*, which is central to the theory of weak derivatives.

Definition 1.6
For $k \in \mathbb{N}$ and $p \in [1,\infty]$, the *Sobolev space of order k*, denoted $W^{k,p}(\Omega)$, is the space of functions for which all the α-th weak derivatives exist and are in $L^p(\Omega)$, for all $|\alpha| \leq k$:

$$W^{k,p}(\Omega) := \{u \in L^p(\Omega) \mid D^\alpha u \in L^p(\Omega),\ |\alpha| \leq k\}.$$

When equipped with the norm

$$\|u\|_{W^{k,p}(\Omega)} := \left(\sum_{|\alpha|\leq k} \|D^\alpha u\|^p_{L^p(\Omega)}\right)^{1/p},$$

the space becomes complete, and thus forms a Banach space.
Recall that $L^p(\Omega)$ is a Banach space for every $p \in [1,\infty]$, but for $p = 2$ it also forms a Hilbert space. We can say this for Sobolev spaces as well. We write $H^k(\Omega) := W^{k,2}(\Omega)$ and equip it with the inner product

$$\langle u, v\rangle_{H^k(\Omega)} := \sum_{|\alpha|\leq k} \langle D^\alpha u, D^\alpha v\rangle_{L^2(\Omega)},$$

forming a Hilbert space. A useful subset of $H^k(\Omega)$ is $H_0^k(\Omega)$, which is defined as the closure of $C_c^\infty(\Omega)$ as a subset of $H^k(\Omega)$. Lastly, we define and $H^{-1}(\Omega)$ as the dual space of $H_0^1(\Omega)$.

When $\partial\Omega$ is sufficiently smooth, we have a more intuitive way to interpret the space $H^1(\Omega)$. In fact, we can say that $H_0^1(\Omega)$ is the subset of all functions $u \in H^1(\Omega)$ that vanish on the boundary, i.e.

$$H_0^1(\Omega) = \{u \in H^1(\Omega) \mid u = 0 \text{ on } \partial\Omega\}.$$

This space is often used for models with homogeneous Dirichlet boundary conditions. The models we will look at mainly use Neumann boundary conditions.

Weak derivatives and Sobolev spaces are generally only used for space variables. The PDEs we will be looking at also include a derivative with respect to time. This is accounted for in a different manner, but it builds upon the last few definitions.

1.3 Variational forms

Using the same method, we can weaken the requirements for solutions to a PDE. Consider, for instance, the heat equation on $\Omega_T := (0,T)\times\Omega$ with homogeneous Dirichlet boundary conditions. For a function u to be a classical solution, we require u to be of class $C^{1,2}(\Omega_T)\cap C^0(\overline{\Omega_T})$, i.e. continuously differentiable with respect to t, twice continuously differentiable with respect to x and continuously extendable to $\overline{\Omega_T}$. We can multiply the heat equation by a test function $\phi\in C_c^\infty(\Omega)$ and integrate on Ω to see

$$\int_\Omega u_t\phi = \int_\Omega \Delta u\phi = \int_{\partial\Omega}\frac{\partial u}{\partial\nu}\phi\,dS - \int_\Omega \nabla u\cdot\nabla\phi = -\int_\Omega \nabla u\cdot\nabla\phi. \tag{1.4}$$

We can also write this as

$$\langle u_t,\phi\rangle = -\langle\nabla u,\nabla\phi\rangle, \tag{1.5}$$

where $\langle\cdot,\cdot\rangle$ denotes the inner product on both $L^2(\Omega)$ and $L^2(\Omega;\mathbb{R}^m)$.

Chapter 2

Turing patterns

Chapter 3

Weak solutions

3.1 Steps

1. For every $v^* \in \mathcal{X}$, define $\mathcal{T}_1(v^*)$ as the solution $(u^*, u_b^*) \in \mathcal{Y}^2$ to the ODEs 3.1 with frozen reactions.
2. For every $(u^*, u_b^*) \in \mathcal{Y}^2$, define $\mathcal{T}_2(u^*, u_b^*)$ as the solution $v^* \in \mathcal{X}$ of the PDE 3.2 with frozen flux.

We want to use the Picard-Lindelöf theorem to prove existence and uniqueness of the ODEs. If we fix a solution of the PDEs v, we have a system

$$\begin{aligned} \partial_t u(t,x) &= f(t,x,u(t,x),u_b(t,x)), \\ \partial_t u_b(t,x) &= g(t,x,u(t,x),u_b(t,x)), \end{aligned} \tag{3.1}$$

where f and g both depend on the trace v. We need f and g to be Lipschitz continuous in u and u_b and continuous in time. The fomer follows from the definition of the model(???), but for the latter, we need continuity of v on ∂C. In other words, for every $x \in \partial C$, we need the function $t \mapsto v|_{\partial C}(t,x)$ to be continuous. Then for every x, we can solve the system 3.1 for $u(\,\cdot\,,x)$ and $u_b(\,\cdot\,,x)$. What can we say about $u(t,\,\cdot\,)$ and $u_b(t,\,\cdot\,)$?

For fixed u and u_b, we have the following equations for v:

$$\begin{aligned} v_t(t,x) &= D\Delta v(t,x) && \text{on } (0,T)\times E, \\ D\frac{\partial v}{\partial \nu}(t,x) &= h(t,x,v(t,x)) && \text{on } (0,T)\times \partial C, \\ D\frac{\partial v}{\partial \nu}(t,x) &= 0 && \text{on } (0,T)\times \partial C, \end{aligned} \tag{3.2}$$

where h depends on u and u_b.

$$\begin{aligned} \mathcal{X} &:= \left\{ v \in L^2(0,T;\Omega) \cap C([0,T];L^2(\Omega)) \mid v \geq 0 \text{ and } v|_{\partial C}(\,\cdot\,,x) \in C([0,T]) \text{ for every } x \in \partial C \right\} \\ \mathcal{Y} &:= \left\{ u \in L^2(0,T;H^1(\Omega)) \mid u \geq 0 \text{ and } u' \in L^2(0,T;H^{-1}(\Omega)) \right\} \end{aligned}$$

Questions:

1. ODE is not Lipschitz, only locally?
2. Picard Lindelöf for weak solutions?
3. Inward flow theorem for weak solutions?
4. When do we have time continuity of the trace?
5. System of ODEs $\rightarrow$ What can we say about $u(t, \cdot)$?

Chapter 4

Simulations

4.1 Problem definitions

4.1.1 Non-diffusing receptors on the cell boundary

This problem assumes the receptor lives on the cell boundaries without diffusing. The ligand is able to diffuse across the entire extracellular space, and interacts with the receptor at the cell boundaries according to the relevant ordinary differential equation. Production of both ligand and receptor exclusively takes place on the cell boundaries.

Let $\Omega \subseteq \mathbb{R}^m$ be the open and bounded domain and denote the cells by an open set $C \subseteq \Omega$ of which the closure $\overline{C}$ is contained in Ω. Furthermore, let $E = \Omega \setminus \overline{C}$ be the extracellular space, and let $l(t,x)$ denote the concentration of ligands at time $t > 0$ and point $x \in \overline{E}$, and $r(t,x)$ the concentration of receptors at time $t > 0$ and point $x \in \partial C$. Then the unknowns l and r should satisfy

$$\begin{aligned} r_t &= \gamma(a - r + r^2 l) && \text{on } (0,\infty) \times \partial C, \\ l_t &= d(???)\Delta l && \text{in } (0,\infty) \times E, \\ \frac{\partial l}{\partial \nu} &= \gamma(b - r^2 l) && \text{on } (0,\infty) \times \partial C, \\ \frac{\partial l}{\partial \nu} &= 0 && \text{on } (0,\infty) \times \partial \Omega, \end{aligned} \tag{4.1}$$

where $d > 0$ is the diffusion coefficient, $\gamma > 0$ is the reaction coefficient and $a > 0$ and $b > 0$ are the production coefficients of the receptor and ligand respectively.

Let $\tau > 0$ be the time discretization parameter and for every $n \in \mathbb{N}$, let $t_n := n\tau$ and

$$r_n(x) \approx r(t_n, x) = r(n\tau, x) \quad \text{and} \quad l_n(x) \approx l(t_n, x) = l(n\tau, x) \tag{4.2}$$

be the approximations of receptor and ligand at time $t_n = n\tau$ respectively. Since we only consider an ODE for r, we do not need to formulate a variational form for this unknown. Instead, we will approximate the ODE for r with some finite difference quotient (???),

$$\frac{r_{n+1} - r_n}{\tau} = \gamma(a - r_n + r_n^2 l_n) \quad \text{on } \partial C, n \in \mathbb{N}, \tag{4.3}$$

using forward Euler. This gives us an explicit formula for r_{n+1},

$$r_{n+1} = r_n + \tau\gamma \left(a - r_n + r_n^2 l_n\right) \quad \text{on } \partial C, n \in \mathbb{N}, \tag{4.4}$$

with r_0 initialized as the exact discretization of the initial conditions on r. To approximate l, we first discretize time using a method slightly different as the one used for r,

$$\frac{l_{n+1} - l_n}{\tau} = d\Delta l_{n+1} \quad \text{in } E, n \in \mathbb{N}. \tag{4.5}$$

The use of the backward Euler instead of forward Euler method provides additional stability, which is particularly convenient considering we are dealing with a second-order derivative. This gives us the implicit formula for l_{n+1},

$$l_{n+1} - \tau d\Delta l_{n+1} = l_n \quad \text{in } E, n \in \mathbb{N}. \tag{4.6}$$

As for r, l_0 is initialized as the exact discretization of the initial condition on l. To derive the variational form of this problem, we multiply both sides by a test function ϕ, integrate on E and use partial integration,

$$\begin{aligned}\int_E l_n\phi &= \int_E [l_{n+1}\phi - \tau d\Delta l_{n+1}\phi] \\ &= \int_E l_{n+1}\phi + \tau d\int_E \nabla l_{n+1}\nabla\phi - \tau d\int_{\partial E}\frac{\partial l_{n+1}}{\partial\nu}\phi\, dS \\ &= \int_E l_{n+1}\phi + \tau d\int_E \nabla l_{n+1}\nabla\phi - \tau d\int_{\partial C}\gamma(b - r_{n+1}^2 l_{n+1})\phi\, dS,\end{aligned} \tag{4.7}$$

where in the last step we used that ∂E is the disjoint union of $\partial\Omega$ and ∂C, since $\overline{C} \subseteq \Omega$, and the Neumann boundary conditions.

$$\int_{\partial C} r_{n+1}\phi = \int_{\partial C}\left[r_n\phi + \tau\gamma\left(a - r_n + r_n^2 l_n\right)\phi\right] \tag{4.8}$$

$u : [0,T] \times \partial C \to \mathbb{R}$ represents receptors, $u_b : [0,T] \times \partial C \to \mathbb{R}$ complex, $v : [0,T] \times E \to \mathbb{R}$ ligand. $k_1, k_2 \in \mathbb{R}$ reaction coefficients, $d_1, d_2 \in \mathbb{R}$ decay coefficients, $D \in \mathbb{R}$ diffusion coefficient, p_1, p_2 some production functions. $(p_1(u) = au,\ p_2(v) = bv?)$

$$\begin{aligned}\partial_t u &= -k_1 u^2 v + k_2 u_b + p_1(u) - d_1 u && \text{on } (0,T) \times \partial C \\ \partial_t u_b &= k_1 u^2 v - k_2 u_b - d_2 u_b && \text{on } (0,T) \times \partial C \\ \partial_t v &= D\Delta v + p_2(v) && \text{on } (0,T) \times E \\ D\partial_\nu v &= -k_1 u^2 v + k_2 u_b && \text{on } [0,T) \times \partial C \\ D\partial_\nu v &= 0 && \text{on } [0,T) \times \partial\Omega\end{aligned}$$